Linear Algebra and Its Applications

Section 5.2: The Characteristic Equation

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.2 The characteristic equation

We use the book's convention $p(\lambda) = \det(A - \lambda I)$. Some software instead reports $\det(\lambda I - A)$; they differ by $(-1)^n$ and have the same zeros. An eigenvalue is a zero of p, and its algebraic multiplicity is its multiplicity as a polynomial root. That multiplicity need not equal the dimension of its eigenspace. For a triangular matrix, the characteristic determinant is the product of the shifted diagonal entries. For a general matrix, use the full determinant rather than multiplying the original diagonal entries.

Practice problem 5.2.P1: A characteristic equation with no real roots

For the given matrix,

$$p(\lambda) = \det \begin{bmatrix} 1-\lambda & -4 \\ 4 & 2-\lambda \end{bmatrix} = (1-\lambda)(2-\lambda) + 16 = \lambda^2 - 3\lambda + 18.$$

Its discriminant is $9 - 72 = -63 < 0$, so the characteristic equation has no real root and A has no real eigenvalue. If complex scalars are allowed, its eigenvalues are $(3 \pm i\sqrt{63})/2$.

Exercise 5.2.1: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (2-\lambda)(2-\lambda) - (7)(7) = \lambda^2 - 4\lambda - 45.$$

Factoring gives $p(\lambda) = (\lambda - 9)(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 9$. Each listed root has multiplicity one.

Exercise 5.2.2: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & -1 \\ 6 & 1 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(1-\lambda) - (-1)(6) = \lambda^2 + 3\lambda + 2.$$

Factoring gives $p(\lambda) = (\lambda + 1)(\lambda + 2)$. Its real roots, hence the eigenvalues, are $-2,\ -1$. Each listed root has multiplicity one.

Exercise 5.2.3: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & 2 \\ 6 & 7 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(7-\lambda) - (2)(6) = \lambda^2 - 3\lambda - 40.$$

Factoring gives $p(\lambda) = (\lambda - 8)(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 8$. Each listed root has multiplicity one.

Exercise 5.2.4: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 8 & 2 \\ 3 & 3 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (8 - \lambda)(3 - \lambda) - (2)(3) = \lambda^2 - 11\lambda + 18.$$

Factoring gives $p(\lambda) = (\lambda - 9)(\lambda - 2)$. Its real roots, hence the eigenvalues, are $2,\ 9$. Each listed root has multiplicity one.

Exercise 5.2.5: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (8 - \lambda)(8 - \lambda) - (4)(4) = \lambda^2 - 16\lambda + 48.$$

Factoring gives $p(\lambda) = (\lambda - 12)(\lambda - 4)$. Its real roots, hence the eigenvalues, are $4,\ 12$. Each listed root has multiplicity one.

Exercise 5.2.6: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 9 & -2 \\ 2 & 5 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (9 - \lambda)(5 - \lambda) - (-2)(2) = \lambda^2 - 14\lambda + 49.$$

Factoring gives $p(\lambda) = (\lambda - 7)^2$. Its real roots, hence the eigenvalues, are 7. The root 7 has multiplicity two.

Exercise 5.2.7: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} 5 & 3 \\ -4 & 4 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (5 - \lambda)(4 - \lambda) - (3)(-4) = \lambda^2 - 9\lambda + 32.$$

The discriminant is $(-9)^2 - 4(32) = -47 < 0$. Consequently there are no real eigenvalues. The two complex roots, if desired, are $(9 \pm i\sqrt{47})/2$.

Exercise 5.2.8: Compute a two-by-two characteristic polynomial

For $A = \begin{bmatrix} -4 & 3 \\ 2 & 1 \end{bmatrix}$, subtract λ only from the diagonal entries. The determinant formula $ad - bc$ gives

$$p(\lambda) = (-\lambda - 4)(1 - \lambda) - (3)(2) = \lambda^2 + 3\lambda - 10.$$

Factoring gives $p(\lambda) = (\lambda - 2)(\lambda + 5)$. Its real roots, hence the eigenvalues, are $-5,\ 2$. Each listed root has multiplicity one.

Exercise 5.2.9: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}4-\lambda & 0 & -1\\ 0 & 4-\lambda & -1\\ 1 & 0 & 2-\lambda\end{bmatrix}.$$

Expand down column two; only its middle entry is nonzero. Thus

$$p(\lambda)=(4-\lambda)[(4-\lambda)(2-\lambda)+1].$$

Multiplying the factors gives

$$\boxed{p(\lambda)=-\lambda^3+10\lambda^2-33\lambda+36}.$$

A useful factored check is $p(\lambda)=-(\lambda-4)(\lambda-3)^2$. Expanding this expression reproduces the same cubic and, at $\lambda=0$, the original determinant.

Exercise 5.2.10: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}3-\lambda & 1 & 1\\ 0 & 5-\lambda & 0\\ -2 & 0 & 7-\lambda\end{bmatrix}.$$

Expand across row two; only its middle entry is nonzero. Thus

$$p(\lambda)=(5-\lambda)[(3-\lambda)(7-\lambda)+2].$$

Multiplying the factors gives

$$\boxed{p(\lambda)=-\lambda^3+15\lambda^2-73\lambda+115}.$$

A useful factored check is $p(\lambda)=-(\lambda-5)(\lambda^2-10\lambda+23)$. Expanding this expression reproduces the same cubic and, at $\lambda=0$, the original determinant.

Exercise 5.2.11: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A-\lambda I=\begin{bmatrix}3-\lambda & 0 & 0\\ 2 & 1-\lambda & 4\\ 1 & 0 & 4-\lambda\end{bmatrix}.$$

Expand down column two; only its middle entry is nonzero, and the remaining minor is triangular. Thus

$$p(\lambda)=(1-\lambda)(3-\lambda)(4-\lambda).$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 8\lambda^2 - 19\lambda + 12}.$$

A useful factored check is $p(\lambda) = -(\lambda - 4)(\lambda - 3)(\lambda - 1)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.12: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} -\lambda - 1 & 0 & 2 \\ 3 & 1 - \lambda & 0 \\ 0 & 1 & 2 - \lambda \end{bmatrix}.$$

Expand across row one. The middle term is zero; the last cofactor contributes $2(3) = 6$. Thus

$$p(\lambda) = (-1 - \lambda)(1 - \lambda)(2 - \lambda) + 6.$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 2\lambda^2 + \lambda + 4}.$$

Factoring is not required here. As checks, the leading term is $-\lambda^3$ under our convention, and substituting $\lambda = 0$ gives $\det A = 4$.

Exercise 5.2.13: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} 6 - \lambda & -2 & 0 \\ -2 & 9 - \lambda & 0 \\ 5 & 8 & 3 - \lambda \end{bmatrix}.$$

Expand down column three; only its last entry is nonzero. Thus

$$p(\lambda) = (3 - \lambda)[(6 - \lambda)(9 - \lambda) - 4].$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 18\lambda^2 - 95\lambda + 150}.$$

A useful factored check is $p(\lambda) = -(\lambda - 10)(\lambda - 5)(\lambda - 3)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.14: Expand the shifted determinant using its zeros

The matrix whose determinant is required is

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 0 & -1 \\ -1 & -\lambda & 4 \\ 0 & 2 & 3-\lambda \end{bmatrix}.$$

Expand across row one. The first minor gives $(-\lambda)(3-\lambda) - 8$, and the last cofactor contributes $(-1)(-2) = 2$. Thus

$$p(\lambda) = (4-\lambda)[(-\lambda)(3-\lambda) - 8] + 2.$$

Multiplying the factors gives

$$\boxed{p(\lambda) = -\lambda^3 + 7\lambda^2 - 4\lambda - 30}.$$

A useful factored check is $p(\lambda) = -(\lambda - 5)(\lambda^2 - 2\lambda - 6)$. Expanding this expression reproduces the same cubic and, at $\lambda = 0$, the original determinant.

Exercise 5.2.15: Read triangular eigenvalues with repetition

This matrix is upper triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = (\lambda - 5)^2 (\lambda - 3)(\lambda - 2).$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list 2, 3, 5, 5. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.16: Read triangular eigenvalues with repetition

This matrix is lower triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = (\lambda - 6)(\lambda - 3)(\lambda - 2)(\lambda + 5).$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list -5, 2, 3, 6. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.17: Read triangular eigenvalues with repetition

This matrix is lower triangular. Therefore $p(\lambda)$ is the product of its diagonal entries minus λ:

$$p(\lambda) = -\lambda (\lambda - 3)^2 (\lambda - 1)^2.$$

Listing the diagonal entries, with each occurrence retained, gives the requested eigenvalue list 0, 1, 1, 3, 3. Repeated entries are repeated eigenvalues in this list, not extra distinct values.

Exercise 5.2.18: Force a two-dimensional eigenspace for four

Subtract $4I$ to obtain

$$A - 4I = \begin{bmatrix} 0 & 2 & 3 & 3 \\ 0 & -2 & h & 3 \\ 0 & 0 & 0 & 14 \\ 0 & 0 & 0 & -2 \end{bmatrix}.$$

The last row of $(A - 4I)x = 0$ forces $x_4 = 0$. The first row then gives $2x_2 + 3x_3 = 0$, so $x_2 = -3x_3/2$. Substituting into the second gives $(h+3)x_3 = 0$. The coordinate x_1 is always free. For the eigenspace to have a second free coordinate, we need $h + 3 = 0$. At $h = -3$ its basis can be chosen as

$$(1, 0, 0, 0)^T, \qquad (0, -3, 2, 0)^T.$$

For any other h, $x_3 = x_2 = x_4 = 0$, leaving only one free coordinate.

Conclusion. $\boxed{h = -3}$.

Exercise 5.2.19: The constant coefficient is the determinant

The supplied factorization is $p(\lambda) = \prod_{j=1}^{n}(\lambda_j - \lambda)$, with eigenvalues repeated according to multiplicity. Substitute $\lambda = 0$:

$$\det A = \det(A - 0I) = p(0) = \lambda_1 \cdots \lambda_n.$$

No sign change is needed under this convention; using $\det(\lambda I - A)$ would introduce the corresponding constant-term sign. The same evaluation works over complex numbers when nonreal roots are present.

Exercise 5.2.20: Transpose preserves the whole characteristic polynomial

Since $I^T = I$ and transposition commutes with subtraction, $A^T - \lambda I = (A - \lambda I)^T$. The determinant of a transpose equals the original determinant. Hence

$$\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$$

for every λ. Thus the polynomials are identical, so the roots agree with their algebraic multiplicities, not merely as distinct sets.

Exercise 5.2.21: Determinants and the sign of a root

(a) **False.** For $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\det A = -1$, but the product of the diagonal entries is 0. The diagonal-product rule is valid for triangular matrices.

(b) **False.** A row interchange negates the determinant; scaling a row by c multiplies it by c. Only row replacement preserves it unchanged in general.

(c) **True.** This is the determinant product law for square matrices.

(d) **False.** A factor $\lambda + 5$ vanishes at $\lambda = -5$, not 5. For example $A = [-5]$ has that factor up to the harmless leading sign, but has only eigenvalue -5.

Exercise 5.2.22: Volume, transpose, multiplicity, and row operations

(a) **False as stated.** The geometric volume is $|\det A|$. The determinant can be negative because it also records orientation.

(b) **False.** $\det A^T = \det A$, with no factor -1. The identity matrix gives 1, not -1.

(c) **True.** This is the definition of algebraic multiplicity.

(d) **False.** Start with $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, whose eigenvalues are $2, 0$. Replacing row two by row two minus row one gives $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, whose eigenvalues are $1, 0$. Row replacement preserves the determinant, not every eigenvalue.

Exercise 5.2.23: Interchanging two factors gives a similar matrix

Assume $A = QR$ with Q invertible. Then

$$Q^{-1}AQ = Q^{-1}(QR)Q = (Q^{-1}Q)RQ = RQ = A_1.$$

Thus $A_1 = Q^{-1}AQ$, precisely the definition of similarity. Only Q needs to be invertible; R need not be. This is why interchanging the factors in the QR iteration preserves eigenvalues.

Exercise 5.2.24: Similarity preserves determinants

If $B = P^{-1}AP$ with P invertible, the product law gives

$$\det B = \det(P^{-1}) \det A \det P.$$

Since $\det(P^{-1}) \det P = \det I = 1$, the two outer factors cancel and $\det B = \det A$. This proof also covers singular A: only P, not A, is required to have an inverse.

Exercise 5.2.25: Resolve a probability vector into two eigenvectors

(a) Find a second eigendirection.

The stated $v_1 = (3/7, 4/7)^T$ satisfies $Av_1 = v_1$. For $v_2 = (1, -1)^T$, multiplication gives $Av_2 = (0.3, -0.3)^T = 0.3v_2$. The vectors correspond to distinct eigenvalues 1 and 0.3, so they are independent and form a basis of $\mathbb{R}^2$.

(b) Expand the initial vector.

$$x_0 - v_1 = (1/2 - 3/7, 1/2 - 4/7)^T = (1/14, -1/14)^T = \tfrac{1}{14}v_2.$$

Thus $x_0 = v_1 + (1/14)v_2$.

(c) Use the eigenvalue powers.

Direct multiplication gives $x_1 = (0.45, 0.55)^T$ and $x_2 = (0.435, 0.565)^T$. For every $k \geq 0$,

$$x_k = A^k x_0 = v_1 + \frac{1}{14}(0.3)^k v_2.$$

Because $|0.3| < 1$, the scalar factor tends to zero, so $x_k \to v_1$. This proves convergence, rather than merely inferring it from two similar iterates.

Exercise 5.2.26: Derive the two-by-two determinant from elimination

Case 1: $a \neq 0$**.**

Replace row two by row two minus (c/a) times row one. This operation does not change the determinant, and it produces $\begin{bmatrix} a & b \\ 0 & d - cb/a \end{bmatrix}$. The triangular formula gives $a(d - cb/a) = ad - bc$.

Case 2: $a = 0$**.**

If also $c = 0$, the first column is zero, so the determinant is zero, agreeing with $ad - bc$. If $c \neq 0$, interchange rows to produce $\begin{bmatrix} c & d \\ 0 & b \end{bmatrix}$. Its determinant is cb, and one interchange reverses the sign, so the original determinant is $-cb = ad - bc$. These cases include all possible values of a, c.

Exercise 5.2.27: Three eigendirections determine a steady-state limit

(a) Check the three given images.

With the matrix in the exercise, direct multiplication gives

$$Av_1 = (0.3, 0.6, 0.1)^T = v_1,$$

$$Av_2 = (0.5, -1.5, 1)^T = \tfrac{1}{2}v_2, \qquad Av_3 = (-0.2, 0, 0.2)^T = \tfrac{1}{5}v_3.$$

All three vectors are nonzero, so these really are eigenvector equations.

(b) Use both a basis and the coordinate sum.

The eigenvalues $1, 1/2, 1/5$ are distinct, so the three eigenvectors are independent and form a basis of $\mathbb{R}^3$. Thus any x_0 has unique coefficients c_j in $x_0 = c_1v_1 + c_2v_2 + c_3v_3$. For $w = (1, 1, 1)^T$, the probability hypothesis gives $w^T x_0 = 1$. But $w^T v_1 = 1$, $w^T v_2 = 0$, and $w^T v_3 = 0$. Applying w^T to the expansion therefore yields $1 = c_1$.

(c) Let the contracting components vanish.

$$x_k = A^k x_0 = v_1 + c_2(1/2)^k v_2 + c_3(1/5)^k v_3.$$

The two powers tend to zero, so $x_k \to v_1$. The two coefficients may depend on the initial distribution, but the limit does not, because its coefficient was fixed by total probability one.

Exercise 5.2.28: Reproducible random examples and the transpose distinction

To make the experiment reproducible, draw upper-triangular entries uniformly from the integers $-3, \ldots, 4$ with pseudorandom seed 52028, setting entries below the diagonal to zero. Successive draws of sizes four and five give the following valid random integer examples. The triangular choice makes exact checks of every multiplicity transparent.

$$A_4 = \begin{bmatrix} -1 & 0 & 0 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Exact calculation gives the same polynomial for A_4 and A_4^T:

$$p(\lambda) = (\lambda - 2)(\lambda - 1)^2(\lambda + 1).$$

Its eigenvalues, with multiplicity, are $-1,\ 1,\ 1,\ 2$. However e_1 is an eigenvector of A_4 with eigenvalue -1, whereas $A_4^T e_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 2 \end{bmatrix}$ is not a multiple of e_1. Thus their eigenvectors are not the same.

$$A_5 = \begin{bmatrix} 1 & 3 & 2 & 3 & 0 \\ 0 & -3 & -3 & 4 & -3 \\ 0 & 0 & 4 & -3 & 4 \\ 0 & 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Exact calculation gives the same polynomial for A_5 and A_5^T:

$$p(\lambda) = -\lambda(\lambda - 4)^2(\lambda - 1)(\lambda + 3).$$

Its eigenvalues, with multiplicity, are $-3,\ 0,\ 1,\ 4,\ 4$. However e_1 is an eigenvector of A_5 with eigenvalue 1, whereas $A_5^T e_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 3 \\ 0 \end{bmatrix}$ is not a multiple of e_1. Thus their eigenvectors are not the same.

The equality of characteristic polynomials is guaranteed in general by Exercise 20; equal eigenspaces are not guaranteed. If a particular matrix is symmetric, its transpose is identical and its eigenspaces do agree, so one must not claim they always differ.

Exercise 5.2.29: Compare a determinant with pivot and eigenvalue products

One reproducible choice of a random integer matrix is

$$A = \begin{bmatrix} 2 & 3 & -3 & -2 \\ 5 & -3 & 0 & 3 \\ -3 & -4 & 0 & 2 \\ 1 & 1 & 4 & 1 \end{bmatrix}.$$

It was selected from pseudorandom entries in $-4, \dots, 5$ (seed 52029) with nonzero pivots and real eigenvalues, so every requested quantity can be displayed in real decimal form.

(a) Eliminate without row scaling.

First use $R_2 \leftarrow R_2 - (5/2)R_1$, $R_3 \leftarrow R_3 + (3/2)R_1$, and $R_4 \leftarrow R_4 - (1/2)R_1$. Continue replacing rows to clear entries below the second and third pivots. No swaps or row scalings are needed. The resulting echelon matrix is

$$U = \begin{bmatrix} 2 & 3 & -3 & -2 \\ 0 & -21/2 & 15/2 & 8 \\ 0 & 0 & -29/7 & -13/21 \\ 0 & 0 & 0 & 74/87 \end{bmatrix}.$$

Row replacement preserves the determinant, so

$$\det A = 2(-21/2)(-29/7)(74/87) = 74.$$

(b) Compute and multiply the eigenvalues.

Exact computation gives $p(\lambda) = \lambda^4 - 40\lambda^2 - 22\lambda + 74$. Its four real roots, computed with guard digits, are approximately

$$-5.8358727568, \quad -1.7435586500, \quad 1.1271321671, \quad 6.4522992397.$$

Their product using unrounded roots is 74 to numerical precision, as also follows exactly from the polynomial's constant coefficient.

(c) Give the requested four-decimal report and comparison.

The pivots are $2.0000, -10.5000, -4.1429, 0.8506$; the eigenvalues are $-5.8359, -1.7436, 1.1271, 6.4523$. The direct matrix determinant is exactly 74, agreeing with both products before rounding. Products formed only from the four-place display may differ slightly because the displayed numbers have been rounded.

Exercise 5.2.30: A small entry change can destroy two real eigenvalues

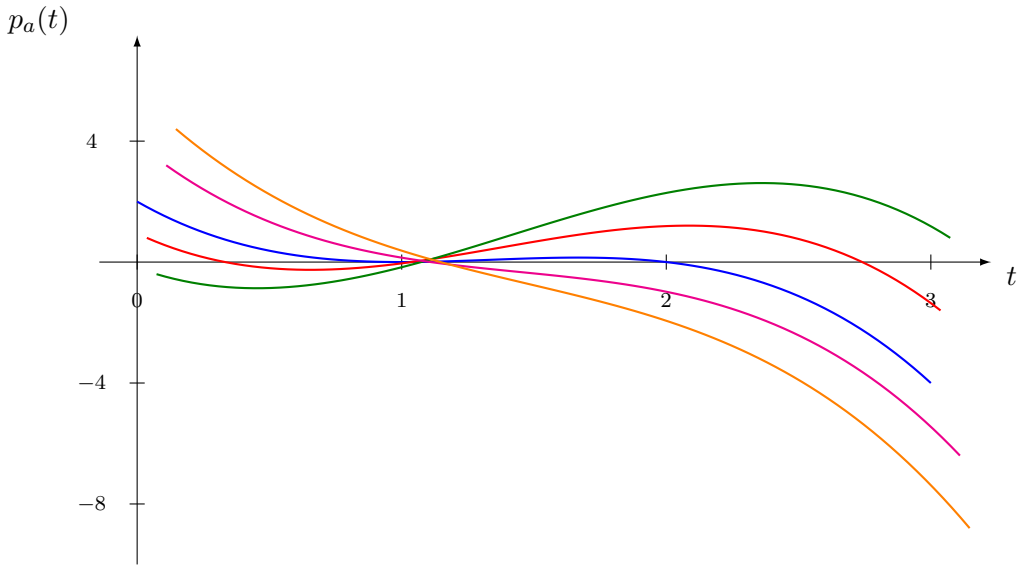
For the stated matrix, expansion of the shifted determinant gives

$$p_a(t) = -t^3 + 4t^2 + (379 - 12a)t + (12a - 382) = -(t-1)(t^2 - 3t + 12a - 382).$$

Thus 1 is always an eigenvalue, and the other two are $(3 \pm \sqrt{1537 - 48a})/2$. Substituting each requested value gives

a	$p_a(t)$	eigenvalues (approximately)
32	$-t^3 + 4t^2 - 5t + 2$	$1, 1, 2$
31.9	$-t^3 + 4t^2 - 3.8t + 0.8$	$0.295841, 1, 2.704159$
31.8	$-t^3 + 4t^2 - 2.6t - 0.4$	$-0.127882, 1, 3.127882$
32.1	$-t^3 + 4t^2 - 6.2t + 3.2$	$1, 1.5 \pm 0.974679i$
32.2	$-t^3 + 4t^2 - 7.4t + 4.4$	$1, 1.5 \pm 1.466288i$

The complex entries in the last two rows are the two nonreal eigenvalues; the only real eigenvalue in those cases is 1. Here are all five graphs on the requested common interval $0 \le t \le 3$.



$a = 32$, $a = 31.9$, $a = 31.8$, $a = 32.1$, $a = 32.2$.

Every graph passes through $(1, 0)$. At $a = 32$, the root 1 is double, so its graph touches the axis there and also crosses at 2. For $a = 31.9$, the three real crossings separate. For $a = 31.8$, two real roots move outside the plotted window; the graph within $[0, 3]$ shows only the crossing at 1, even though all three roots remain real. For $a = 32.1, 32.2$, the quadratic discriminant is negative, so those two roots are genuinely nonreal, not merely off screen. The coalescence threshold for that pair is $a = 1537/48 \approx 32.020833$.

Conclusion. The exact factorization explains the changes. A graph on a limited window alone cannot distinguish a root outside the window from a nonreal root; the discriminant and computed values supply that distinction.